Renewable Energy in the Indian Subcontinent: A Comparative Review with Focus on India

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Abstract—The Indian subcontinent is home to one of the world's most significant and diverse renewable energy transitions. This paper presents a comprehensive comparative analysis of renewable energy system (RES) deployment across seven South Asian nations — India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives — with particular focus on India's dominant role as the region's leading renewable energy producer. Drawing on installed capacity data, policy frameworks, investment trends, and socioeconomic impact assessments, the study examines solar, wind, hydro, biomass, and emerging ocean energy systems. A SWOT analysis framework is applied to identify each country's strategic position. Empirical regression models relating solar capacity to land area and capital cost are presented for India and Pakistan. Results indicate that India's total installed renewable capacity reached approximately 152.43 GW by 2021–22, representing 38% of its total power generation mix. The study identifies untapped tidal energy potential across the subcontinent as a critical unaddressed opportunity, and highlights governance, infrastructure, and investment barriers that constrain energy transition in smaller nations. Findings support evidence-based policy recommendations for accelerating renewable adoption across the region.

Index Terms—*renewable energy, solar energy, wind energy, Indian subcontinent, SWOT analysis, energy policy, comparative analysis, South Asia.*

I. INTRODUCTION

The global shift towards sustainable and clean energy has gained considerable momentum over the past two decades. Countries across the world are under increasing pressure to reduce their dependence on fossil fuels, mitigate greenhouse gas emissions, and ensure long-term energy security [1]. Nowhere is this challenge more acute — or more consequential — than in the Indian subcontinent, a region that collectively houses approximately 1.9 billion people and accounts for a substantial and growing share of global energy demand.

The Indian subcontinent encompasses seven nations — India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives — each with distinct geographical characteristics, climatic conditions, economic structures, and energy needs. This diversity presents both extraordinary opportunities and significant challenges for renewable energy development [2]. South Asia's varied topography, ranging from the Himalayan highlands to tropical coastlines, supports a rich portfolio of renewable resources, including solar irradiation, wind corridors, hydroelectric potential, and biomass reserves.

India stands as the region's clear frontrunner, ranking third globally in total renewable energy capacity and third in Ernst and Young's Renewable Energy Country Attractiveness Index (RECAI) in 2021 [3]. By 2020, renewables accounted for 38% of India's total installed power capacity of 373 GW. India has committed under the Paris Agreement to generate 50% of its electricity from non-fossil fuel sources by 2030 and has set an ambitious target of 500 GW of renewable capacity by that year.

Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives each occupy distinct positions on the renewable energy development spectrum. Pakistan possesses enormous untapped solar and wind potential but is constrained by political instability and infrastructure deficits [4]. Bangladesh has achieved remarkable off-grid solar success but faces grid integration challenges [5]. Sri Lanka aims for energy independence by 2030, while Nepal and Bhutan leverage their substantial hydropower resources. The Maldives, heavily dependent on imported diesel, is pioneering hybrid island energy systems.

This paper synthesises the current status, policy landscape, and socioeconomic impacts of renewable energy across all seven nations, with India as the primary reference point. The objectives of this study are: (i) to document and compare installed renewable energy capacities across the subcontinent; (ii) to assess the role of RES in driving economic growth, investment, and environmental sustainability; (iii) to apply SWOT analysis to each country's renewable energy sector; and (iv) to identify regional gaps and policy recommendations, with particular attention to unexploited tidal and ocean energy potential.

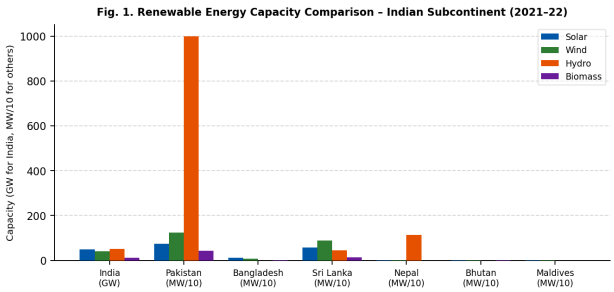

Fig. 1. Renewable Energy Capacity Comparison — Indian Subcontinent (2021–22)

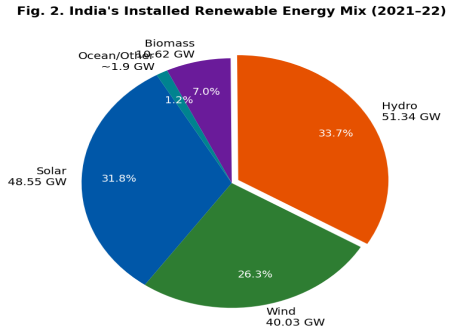

Fig. 2. India's Installed Renewable Energy Mix (2021–22)

II. METHODOLOGY

This study employs a structured narrative review methodology supplemented by descriptive statistical analysis and empirical regression modelling. The research framework follows a four-stage process as illustrated in Fig. 3.

The literature review phase involved a systematic search of peer-reviewed publications and institutional reports from 2000 to 2022, sourced from Scopus, Web of Science, Google Scholar, and official repositories including the International Renewable Energy Agency (IRENA), International Energy Agency (IEA), and national energy ministries. Inclusion criteria required relevance to renewable energy in at least one of the seven subcontinent nations, with empirical data on capacity, policy, or impact outcomes.

Quantitative data on installed capacities, project characteristics, and investment flows were compiled into a unified comparative dataset. Descriptive statistics including mean, standard deviation, and range were computed for state-wise and country-wise capacity data. Two regression models, derived from solar project datasets for India (Eq. 1) and Pakistan (Eq. 2), relate per-MW solar capacity to land area (acres) and capital cost (USD million):

$$1\ MW\ (India) = 0.0662 \times Area + 1.0193 \times Cost - 229.03 \quad (1)$$

$$1\ MW\ (Pakistan) = 0.0270 \times Area + 0.3497 \times Cost - 22.64 \quad (2)$$

SWOT analysis was applied to each country using evidence drawn from the reviewed literature, with factors categorised into internal (Strengths, Weaknesses) and external (Opportunities, Threats) dimensions.

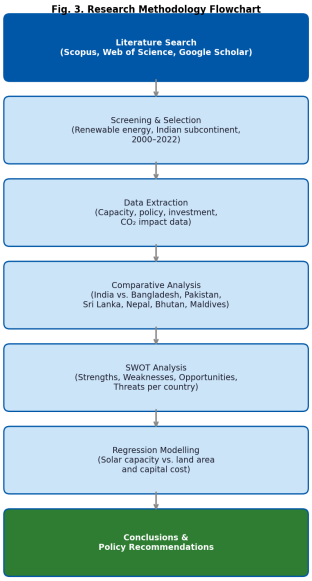

Fig. 3. Research Methodology Flowchart

III. RENEWABLE ENERGY STATUS: INDIA

India occupies a commanding position in the global renewable energy landscape. As of 2021–22, the country's installed renewable capacity stood at approximately 152.43 GW, distributed across solar (48.55 GW), wind (40.03 GW), hydro (51.34 GW), biomass (10.62 GW), and ocean/other sources (~1.9 GW) [3]. This positions India as the world's third-largest renewable energy producer, behind only China and the United States.

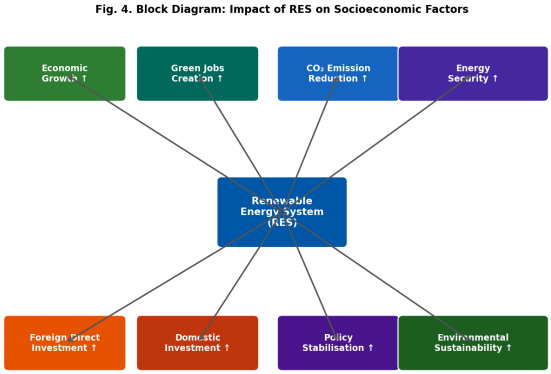

Fig. 4. Block Diagram: Impact of RES on Socioeconomic Factors

A. Solar Energy

Solar energy represents one of India's most dynamic and rapidly expanding renewable sectors. The country's solar irradiation varies from 3.5 to 7.0 kWh/m²/day across its states, with maximum radiation recorded in Madhya Pradesh, Chhattisgarh, Haryana, Punjab, and Rajasthan [6]. States receiving more than 6 kWh/m²/day are considered highly suitable for utility-scale solar deployment.

By September 2020, India had operationalised 89.22 GW of solar capacity, with a further 48.21 GW under execution and 25.64 GW in bidding stages. India hosted three of the world's five largest solar parks: the 2,245 MW Bhadla Solar Park in Rajasthan, the 2,050 MW Pavagada Solar Park in Karnataka, and the 1,000 MW Kurnool Ultra Mega Solar Park in Andhra Pradesh. Table II presents the top five solar parks by capacity.

B. Wind Energy

India's wind sector is supported by a robust domestic manufacturing base comprising 20 companies producing 53 internationally certified turbine models with capacities ranging from 3 to 30 MW. Wind turbines manufactured in India are exported to Europe, the United States, and other markets, underscoring the sector's industrial maturity [7]. Tamil Nadu leads all states with 9,231 MW of installed wind capacity, followed by Rajasthan (4,300 MW) and Maharashtra (4,794 MW). India's RECAI score for onshore wind stands at 54.2 (rank 6 globally), though offshore wind capacity remains nascent (rank 29, score 28.6).

C. Hydro and Biomass

Hydropower represents India's single largest renewable source at 51.34 GW. Arunachal Pradesh alone accounts for 52,128 MW of hydropower potential. Biomass capacity totals 10.62 GW nationally, led by Maharashtra (2,632 MW) and Uttar Pradesh (2,180 MW). India's shift to a green economy is projected to generate up to 3 million renewable energy jobs by 2030, with 432,000 already employed in the sector as of 2017 [8].

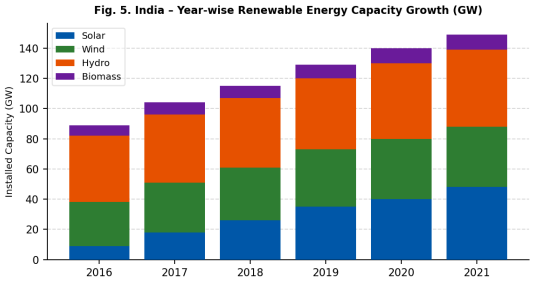

Fig. 5. India – Year-wise RE Capacity Growth (GW)

D. Policy and Investment

India's policy framework for renewable energy is among the most comprehensive in the developing world. Key initiatives include the National Solar Mission (targeting 100 GW of solar by 2022), Production Linked Incentive (PLI) schemes for solar module manufacturing, net metering regulations, and international partnerships including the India-USA Strategic Energy Partnership and the International Solar Alliance. In the first half of 2018 alone, new investments in non-conventional energy totalled USD 7.4 billion, and the sector has attracted USD 7.83 billion in FDI equity over the preceding decade [9].

IV. COMPARATIVE ANALYSIS: OTHER SUBCONTINENT NATIONS

A. Pakistan

Pakistan's renewable energy sector, though still nascent, holds enormous potential. The country's Alternative Renewable Energy Policy targets 20% renewable contribution to the national grid by 2025 and 30% by 2030. Renewables currently account for approximately 7% of installed capacity, with hydropower as the dominant source (10,002 MW). Solar irradiation averages 5.3 kWh/m²/day, and wind potential in Sindh and Baluchistan is estimated at 43,000 MW across 9,700 km². Pakistan's first 50 MW wind plant was commissioned in 2013, and the government has since moved to competitive/reverse bidding for solar and wind projects [4]. However, political instability, poor grid infrastructure, and lack of skilled manpower remain critical barriers.

B. Bangladesh

Bangladesh is distinguished by its world-class off-grid solar programme, widely recognised by the World Bank as one of the most effective of its kind globally. The country leads Asia alongside China, India, and Japan in renewable energy development metrics [5]. However, its grid-connected renewable capacity remains limited: as of 2022, planned capacity totals only 215 MW across solar, wind, biomass, and biogas. Key constraints include limited onshore wind potential, high biomass collection costs, a lack of incentives for private investors, and insufficient coordination between ministries.

C. Sri Lanka

Sri Lanka aims to achieve energy independence by 2030, targeting a generation capacity of 6,900 MW — up from 4,043 MW — with substantial renewable additions including 1,389 MW solar, 1,205 MW wind, and 842 MW major hydro between 2018 and 2037. The country already achieves a 98% grid connection rate by South Asian standards, a remarkable achievement. Annual electricity consumption growth of approximately 4.9% per year is projected over the next two decades, straining both conventional and renewable supply [10].

D. Nepal, Bhutan, and the Maldives

Nepal's primary renewable resource is hydroelectricity, with a potential of over 83 GW, of which only 1,142 MW is currently operational. Average solar radiation of 4.7 kWh/m²/day over approximately 300 sunny days annually offers complementary potential. The country's December 2017 wind-solar hybrid launch in the Sindhuli district marked a significant milestone for rural electrification [11].

Bhutan, as a carbon-negative nation, derives 99% of its electricity from hydropower. Its total hydropower capacity of 1,614 MW significantly exceeds domestic demand, enabling substantial export to India. The 12th Five-Year Plan (2018–2023) prioritises pilot renewable projects beyond hydro, including small solar and wind installations. Electric taxis are being introduced in urban centres.

The Maldives became the first nation in the world to achieve universal electricity access in 2008. However, approximately 94% of its generation capacity (272 MW) remains diesel-based, resulting in among the highest power generation costs in South Asia. Daily solar irradiation of 4.5 to 6 kWh/m²/day and an annual solar energy potential of 400 million MW represent transformative untapped resources. Studies indicate a fully renewable system — utilising floating solar PV and wave energy converters — could theoretically be achieved by 2030 [12].

V. SOCIOECONOMIC IMPACT OF RENEWABLE ENERGY

As illustrated in Fig. 4, the adoption of renewable energy systems exerts wide-ranging positive effects on the socioeconomic fabric of subcontinent nations, extending well beyond electricity generation. The relationship between renewable energy consumption and key macroeconomic indicators can be expressed as:

$$ln(RE) = \beta_0 + \beta_1(EG) + \beta_2(ES) + \beta_3(FI) + \beta_4(JP) + \beta_5(SQL) \quad (3)$$

where RE is renewable energy consumption, EG is economic growth, ES is environmental sustainability, FI is foreign investment, JP is job prospects, and SQL is sub-optimal quality of life [3].

For India, the transition to a green economy is expected to create 3 million jobs in the renewable sector alone by 2030. As of 2017, over 721,000 Indians were already employed in green sectors, accounting for more than 20% of all new green jobs created globally that year. Renewable energy investment has also catalysed significant GDP growth: new investments in India's non-conventional energy sector totalled USD 7.4 billion in the first half of 2018 [8].

In Pakistan, the most distinctive socioeconomic effect identified in the literature is terrorism reduction — decentralised energy systems in rural areas reduce the economic marginalisation that feeds extremism. In Bhutan, environmental quality and service sector growth are the primary observed benefits. For Bangladesh,

the primary economic co-benefits are CO_2 emission reductions and improved foreign direct investment climates. Sri Lanka's renewable energy deployment has attracted foreign capital and stabilised long-term energy policy frameworks [10].

VI. SWOT ANALYSIS

Table III presents a consolidated SWOT analysis for renewable energy systems across the Indian subcontinent, structured to highlight the contrast between India's maturity and the comparative position of other nations.

TABLE III. SWOT ANALYSIS — RENEWABLE ENERGY SYSTEMS IN THE INDIAN SUBCONTINENT

Factor	India	Comparator Nations
Strengths	3rd largest RE producer globally RECAI rank 3 (EY 2021) Strong solar/wind manufacturing base 38% RE of total installed capacity	Pakistan: govt. incentives, tax exemptions Sri Lanka: 98% grid connectivity Bangladesh: top 4 in Asia RE development Nepal/Bhutan: carbon-negative, hydro surplus
Weaknesses	Intermittent weather conditions Varied tariff rates across states High capital cost barrier Skilled manpower shortage	Pakistan: poor grid infrastructure Bangladesh: limited wind potential onshore Maldives: only 6% RE share (2020) Nepal: underdeveloped transmission grid
Opportunities	Tidal energy at Gulf of Kutch/Khambhat Offshore wind expansion Geothermal potential Green hydrogen production	Pakistan: 60 GW hydropower potential Bangladesh: offshore wind unexplored Maldives: floating solar + wave energy Nepal: wind-hydro seasonal complementarity
Threats	Inefficiency in risk analysis Mitigation mechanism gaps Land acquisition challenges	Pakistan: terrorism, political instability Bangladesh/Sri Lanka: inadequate infrastructure Maldives: land area constraints Nepal/Bhutan: high LCOE for remote sites

VII. CAPACITY COMPARISON AND TABULAR SUMMARY

Table I presents the consolidated renewable energy installed capacities across all seven nations as of 2021–22. India's dominance is clear: its solar capacity alone (48.55 GW) exceeds the entire renewable portfolio of every other nation on the subcontinent. Pakistan's hydropower capacity (10,002 MW) is the strongest among the comparator group, reflecting the country's reliance on large hydro projects in its northern regions.

TABLE I. COMPARATIVE RENEWABLE ENERGY INSTALLED CAPACITY — INDIAN SUBCONTINENT (2021–22)

Country	Solar	Wind	Hydro	Biomass	Total RE	Unit
India	48.55	40.03	51.34	10.62	152.43	GW
Pakistan	737	1,236	10,002	432	12,407	MW
Bangladesh	120	80	0	15	215	MW
Sri Lanka	563	887	438	129	2,017	MW
Nepal	5	3	1,142	—	1,150	MW
Bhutan	6.1	5.1	1,614	1	1,626	MW
Maldives	15	1	0	0	16	MW

TABLE II. TOP FIVE SOLAR POWER PARKS IN INDIA BY INSTALLED CAPACITY

S.N.	Solar Park / Project	State	Capacity (MW)	Area (Acres)	Capital Cost (USD M)
1	Bhadla Solar Park	Rajasthan	2,245	14,000	1,428.57
2	Pavagada Solar Park	Karnataka	2,050	13,000	1,285.71
3	Kurnool Ultra Mega Solar Park	Andhra Pradesh	1,000	5,932	1,000.00
4	Rewa Ultra Mega Solar	Madhya Pradesh	750	1,590	642.86
5	Charanka Solar Park	Gujarat	615	4,900	766.43

VIII. UNTAPPED POTENTIAL: TIDAL AND OCEAN ENERGY

A notable finding of this comparative study is that no nation in the Indian subcontinent currently exploits tidal energy for electricity generation, despite several locations possessing significant tidal ranges. In India, the Gulf of Kutch and Gulf of Khambhat in Gujarat, the Palk Bay and Mannar Channel in Tamil Nadu, the Hooghly River, and the Sundarbans in West Bengal all offer viable tidal power sites. Ocean energy attractiveness scores for India indicate considerable unexplored potential beyond the current RECAI score positions [3].

Bangladesh hosts multiple high-potential tidal sites including Sandwip, Cox's Bazar, Hiron Points, and Golachippa. The Maldives — with tidal ranges exceeding 6–7 m at Cocoa Island, Veligandu Island, Thulusdhoo Island, Hulhumale Island, and Fulhadhoo Island — possesses resources that could substantially reduce its dependence on imported diesel. Floating offshore technologies combining solar PV and wave energy converters have been identified as technically feasible for full decarbonisation of Maldivian islands by 2030 [12].

The absence of operational tidal projects across the entire subcontinent represents a missed opportunity, particularly for island and coastal nations where tidal energy could provide stable baseload generation that complements the intermittency of solar and wind. Policy instruments to support feasibility studies, pilot projects, and technology transfer for ocean energy should be prioritised by regional energy ministries.

IX. DISCUSSION

The analysis presented in this paper reveals a subcontinent at markedly different stages of renewable energy maturity, with a single dominant player — India — and a group of nations whose energy transitions are still in formative phases. Several cross-cutting observations warrant discussion.

First, hydropower remains the backbone of renewable energy in all seven nations when measured by absolute capacity. However, the solar energy revolution underway in India — driven by rapidly falling PV costs, ambitious national targets, and innovative auction mechanisms — is reshaping the regional energy map. India's solar capacity has grown from under 10 GW in 2016 to over 48 GW in 2021, a pace with few global parallels.

Second, policy architecture is the single most differentiating factor between nations. India's sophisticated blend of feed-in tariffs, competitive auctions, PLI schemes, net metering, and international investment partnerships has mobilised capital at scale. Pakistan, despite comparable resource endowments, lacks the institutional stability and grid infrastructure to replicate this success. Bangladesh's off-grid success demonstrates that context-appropriate technology deployment can achieve remarkable outcomes even without robust national grid infrastructure.

Third, the socioeconomic co-benefits of renewable energy — job creation, GDP growth, poverty alleviation, and, notably in the case of Pakistan, security improvement — strengthen the case for investment beyond traditional energy security and emissions arguments. Policymakers across the subcontinent should explicitly recognise and quantify these co-benefits in national planning frameworks.

Fourth, the regression equations derived for India (Eq. 1) and Pakistan (Eq. 2) offer a practical tool for pre-feasibility assessment of solar projects, enabling rapid estimation of required land and capital for a given capacity target. These relationships should be validated with updated datasets as project costs continue to decline.

X. CONCLUSION

This paper has presented a comprehensive comparative analysis of renewable energy system status across the seven nations of the Indian subcontinent, with India as the primary reference point. The key findings are as follows:

- India dominates the subcontinent's renewable energy landscape with 152.43 GW of installed capacity, representing 38% of its total power mix, and is on track for its 500 GW target by 2030.
- Pakistan's renewable sector remains constrained by political instability and infrastructure gaps, despite significant solar and wind resource potential.
- Bangladesh's off-grid solar programme is a global model; however, grid-connected renewable capacity remains extremely limited.
- Nepal and Bhutan possess strong hydropower resources; the Maldives urgently requires a transition from diesel-based generation to renewable hybrid systems.
- Tidal and ocean energy potential across the subcontinent is substantial but entirely unexploited, representing a critical gap for future policy and investment.
- SWOT analysis reveals that governance quality, infrastructure investment, and technology transfer are the primary determinants of renewable energy transition success across the region.

Future research should address the statistical validation of capacity-cost regression models, systematic tidal resource mapping across coastal subcontinent nations, and multi-country energy transition modelling incorporating demand growth projections to 2050.

. REFERENCES

[1] V. Khare, S. Nema, and P. Baredar, "Solar-wind hybrid renewable energy system: a review," Renew. Sustain. Energy Rev., vol. 58, pp. 23–33, 2016.

[2] V. Khare, C. J. Khare, S. Nema, and P. Baredar, "Path towards sustainable energy development: Status of renewable energy in Indian subcontinent," Cleaner Energy Systems, vol. 3, p. 100020, 2022.

[3] A. Das, "A comprehensive review of wind–solar hybrid energy policies in India," Renew. Energy Focus, vol. 35, pp. 108–121, 2020.

[4] M. Kamran, "Towards empowerment of the renewable energy sector in Pakistan for sustainable energy evolution: SWOT analysis," Renew. Energy, vol. 146, pp. 543–558, 2020.

[5] M. A. H. Baky, "Development of renewable energy sector in Bangladesh: current status and future potentials," Renew. Sustain. Energy Rev., vol. 73, pp. 1184–1197, 2017.

[6] J. Shekhar, "Reduced renewable energy stability in India following COVID-19," Renew. Sustain. Energy Rev., vol. 144, p. 111015, 2021.

[7] S. K. Pathak, "Prioritization of barriers to the development of renewable energy technologies in India," Sustain. Energy Technol. Assess., vol. 50, p. 101818, 2022.

[8] D. Deshwal, "How will COVID-19 impact renewable energy in India?" Energy Res. Soc. Sci., vol. 77, p. 102097, 2021.

[9] Q. Wang, "India's renewable energy: new insights from multi-regional input output and structural decomposition analysis," J. Clean. Prod., vol. 283, p. 124230, 2021.

[10] C. H. Dasanayaka et al., "Investigating the effects of renewable energy utilization towards the economic growth of Sri Lanka," Clean. Eng. Technol., vol. 6, p. 100377, 2022.

[11] A. Suman, "Role of renewable energy technologies in climate change adaptation and mitigation: a brief review from Nepal," Renew. Sustain. Energy Rev., vol. 151, p. 111524, 2021.

[12] D. Keiner et al., "Powering an island energy system by offshore floating technologies towards 100% renewables: a case for the Maldives," Appl. Energy, vol. 308, p. 118360, 2022.